Department: Computer Science and Engineering

Semester: 7

Capstone Project – I

Vita Watch: Healthcare Alert System

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Introduction of the Research domain

- Healthcare monitoring and risk prediction, a crucial aspect of modern medicine, addresses conditions influenced by factors such as age, lifestyle, genetics, and underlying health issues (e.g., high blood pressure, respiratory anomalies). Early identification and timely alerts are vital for mitigating risks, but traditional approaches often lack real-time insights and accessibility.

- This research domain focuses on leveraging machine learning and data analytics to develop a Healthcare Risk Prediction System that evaluates the likelihood of health risks based on vital sign data. By analysing factors such as temperature, heart rate, and oxygen saturation, the system predicts risk levels and offers personalised recommendations for intervention and prevention. This approach aims to provide an accessible, user-friendly platform that enables proactive health management and supports timely medical decision-making.

Literature Review

Sr. No.	Paper Title	Publication	Year
1.	Health Risk Assessment Using Machine Learning: Systematic Review	MDPI: electronics	2022

- Objective:- To develop and evaluate machine learning models for predicting healthcare risks using vital sign data, enabling early detection and intervention based on real-time health assessments.

- Research outcomes:- The study used algorithms like Logistic Regression, Decision Trees, and Random Forest, achieving 90% accuracy with Random Forest. Key predictors identified include heart rate, temperature, blood pressure, and oxygen saturation.

Literature Review(Cont..)

Sr. No.	Paper Title	Publication	Year
2.	A Catalogue of Machine Learning Algorithms for Healthcare Risk Prediction	MDPI: Sensors	2022

- Objective:- To analyze vital sign data to identify and rank the key healthcare risk factors.

- Research outcomes:- Data analysis revealed vital signs like heart rate, temperature, and blood pressure significantly impact risk levels. The integration of real-time wearable device data enhances monitoring accuracy. A predictive model was proposed to improve personalized healthcare risk assessment.

Comparative Analysis

Table 1: Comparative Analysis Table

Sr No.	Paper Title	Model/Method	Finding/Findings
1.	Health Risk Assessment Using Machine Learning: Systematic Review	• Systematic review of various ML models (e.g., SVM, Random Forest) applied to healthcare data.	• Random Forest and SVM are effective; importance of feature selection and explainability; challenges include data imbalance and privacy.
2.	A Catalogue of Machine Learning Algorithms for Healthcare Risk Prediction	• Catalogue comparing models (e.g., Naïve Bayes, KNN, Random Forest, Neural Networks) for specific healthcare tasks.	• Random Forest and Neural Networks perform well; challenges with data preprocessing and model adaptation.

Identified Research Gap

1. **Dataset Integration:** There's a gap in integrating various healthcare datasets (like EHRs, imaging, and genomic data) for better classification accuracy. This integration can lead to more comprehensive models that provide personalized healthcare predictions.

2. **Real-Time Processing:** Research is needed to focus on the real-time processing of healthcare data for timely interventions. Developing models that handle streaming data can support clinical workflows and improve patient care.

3. **Model Explainability:** Ensuring the explainability and transparency of machine learning models in healthcare is crucial. This helps healthcare professionals understand and trust AI-driven predictions, fostering better adoption in clinical practice.

Problem Statement

- Despite advancements in healthcare technology, accurately predicting patient health risks remains a challenge due to the complexity and variability of medical data. Existing models often fail to integrate diverse health metrics, real-time data, and personalized factors, resulting in limited predictive accuracy and effectiveness in clinical settings. This gap hampers timely intervention and personalized patient care, leading to suboptimal health outcomes.

- Additionally, current predictive models lack transparency and explainability, making it difficult for healthcare professionals to trust and rely on their predictions. There is an urgent need for a comprehensive healthcare classification system that leverages advanced machine learning techniques to analyze and predict health risks. Such a system should incorporate diverse data sources, provide real-time predictions, and offer clear, actionable insights to support personalized patient care and improve overall health outcomes.

Objectives of the Work

1. Develop a robust machine learning model to classify patient health risks using a diverse healthcare dataset.
2. Enhance model transparency and interpretability for healthcare professionals.
3. Utilize a variety of healthcare data sources for comprehensive analysis.
4. Ensure the classification system is scalable and flexible.
5. Validate the model's performance through thorough evaluation metrics and real-world testing.

Proposed Methodology

1. **Data Collection:** Gather clinical and lifestyle data from publicly available datasets.
2. **Data Preprocessing:** Clean and preprocess data to handle missing values, outliers, and normalize features for better model performance.
3. **Feature Selection:** Identify key features influencing patient health risks using statistical methods and domain expertise.
4. **Model Development:** Develop predictive model using Random Forest algorithm in Machine Learning.
5. **Model Evaluation:** Evaluate models using performance metrics like accuracy, precision, recall, and F1-score to select the best-performing model.
6. **Personalized Guidance system:** Design a user-friendly interface providing actionable health recommendations based on individual risk profiles.
7. **Validation and Testing:** Validate the system using new data and test it.

Implementation Work

1. **Data Preparation:** Collected and preprocessed datasets, ensuring data quality and normalization for machine learning.
2. **Model Training and Testing:** Developed predictive models using the algorithm Random Forest.
3. **Feature Analysis:** Performed feature selection to identify key factors influencing patient health risks.
4. **System Integration:** Integrate the predictive model into a user-friendly Dash application for easy interaction and visualization of health risk predictions.
5. **Validation:** Evaluated the system with new data to ensure accuracy and adaptability for various patient groups.

Timeline Chart

1. **Month 1:** Data collection and preprocessing. Identifying and preparing datasets for analysis.
2. **Month 2:** Feature selection and exploratory data analysis. Development of predictive models using machine learning and optimization.
3. **Month 3:** Development of the Dash application and designing user-friendly interfaces for health risk predictions.
4. **Month 4:** System testing, validation, and fine tuning. Preparing documentation and final presentation.

References

1. Machine Learning model that healthcare providers will use

2. Machine Learning (ML) in Medicine: Review, Application and Challenges

3. Kaggle

4. Copilot